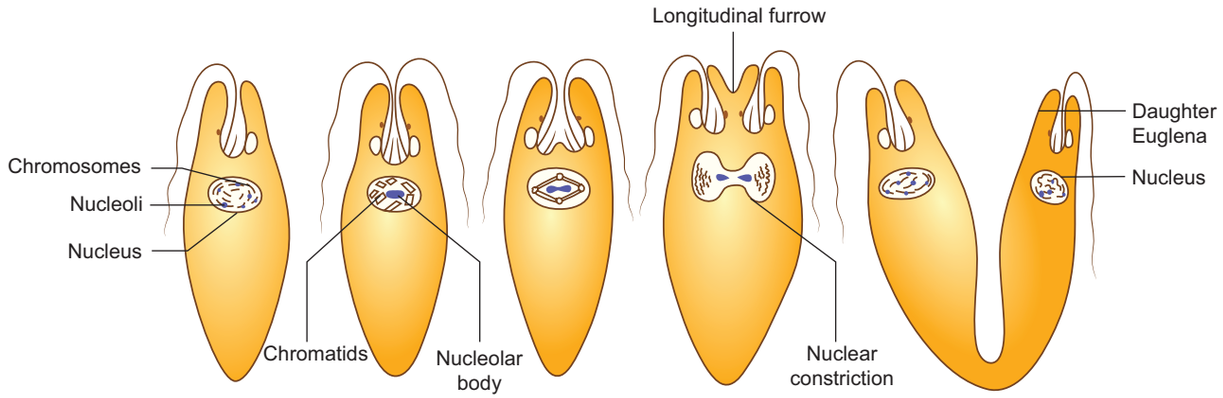


Fig. 1.3 Longitudinal binary fission in *Euglena*

flagellates, the flagellum is retained usually by one daughter cell. The basal granule is divided into two and the new basal granule forms a flagellum in the other daughter individual. e.g. *Vorticella* and *Euglena.*

In **oblique binary fission** the plane of division is oblique. It is seen in dinoflagellates. e.g. *Ceratium.*

In **multiple fission** the parent body divides into many similar daughter cells simultaneously. First, the nucleus divides repeatedly, later the cytoplasm divides into as many parts as that of nuclei. Each cytoplasmic part encircles one daughter nucleus. This results in the formation of many smaller individuals from a single parent organism. If multiple fission produces four or many daughter individuals by equal cell division and the young ones do not separate until the process is complete, then this division is called **repeated fission.** e.g. *Vorticella*.

During unfavorable conditions (increase or decrease in temperature, scarcity of food) *Amoeba* withdraws its pseudopodia and secretes a three-layered, protective, chitinous cyst wall around it and becomes inactive (**Fig. 1.4**). This phenomenon is called **encystment.** When conditions become favourable, the encysted *Amoeba* divides by multiple fission and produces many minute amoebae called **pseudopodiospore** or **amoebulae.** The cyst wall absorbs water and breaks off liberating the young pseudopodiospores, each with a fine pseudopodia. They feed and grow rapidly to lead an independent life.

In some metazoan animals, a special type of transverse fission called **strobilation** occurs

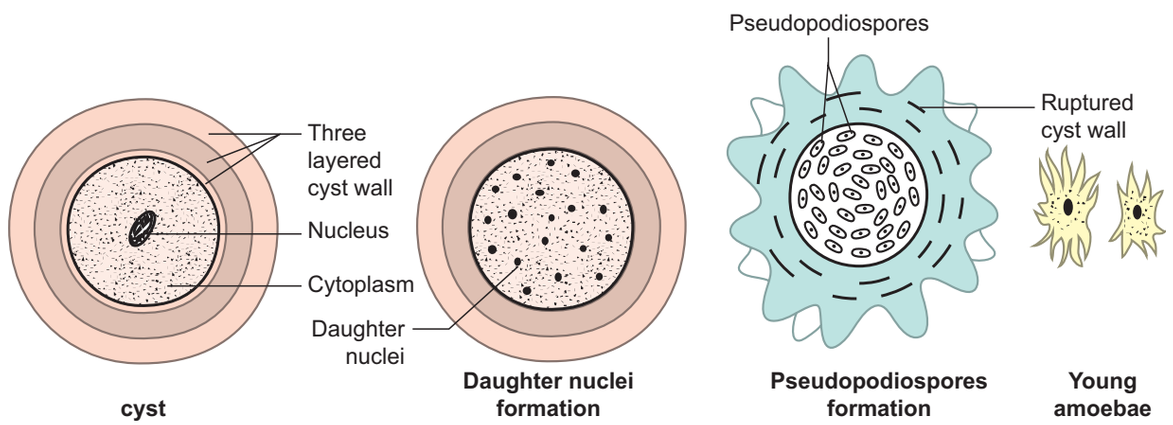


Fig. 1.4 Multiple fission in encysted *Amoeba*

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